IGOR Final Report

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Executive Summary

This research initiative investigated the root causes of suppressed protein yield in the PURE cell-free system when executed on specific Labcraft hardware. The IGOR process, integrating literature analysis, AI-driven hypothesis generation, and rigorous debate, successfully pivoted the investigation from a purely biochemical optimization problem to a more comprehensive biophysical one. The key outcome was the identification and prioritization of two evolved hypotheses that link the hardware’s physical properties to known biochemical failure modes. The winning hypothesis, ‘Synergistic Biophysical Failure’, posits that a combination of accelerated evaporation and surface-mediated sequestration of inhibitory complexes is responsible for the yield loss. The subsequent experimental plan moves beyond simple component titration to a multivariate, combinatorial design aimed at de-convoluting these physical effects, thereby guiding the next phase of optimization for this specific hardware platform.

Top-Ranked Hypotheses

1: Synergistic Biophysical Failure: A Combined Model of Evaporation-Driven Concentration and Surface-Mediated Sequestration

Hypothesis: This hypothesis posits that the severe yield loss on Labcraft hardware results from two synergistic physical failures. First, accelerated evaporation non-linearly concentrates solutes, increasing the kinetic probability of forming inhibitory magnesium-phosphate (Mg-P) complexes. Second, the specific surface properties of the Labcraft hardware preferentially adsorb these newly-formed inhibitory complexes. This creates localized, diffusion-limited depletion zones of essential free Mg near reaction surfaces, leading to a more rapid and catastrophic reaction stall than either mechanism could achieve alone. This unified model explains why simple biochemical re-optimization fails and suggests that a solution must address both the bulk concentration effects and the surface-interface interactions. **Final Rationale:** Evolved from 'Architect 1' by incorporating the core mechanism of 'Explorer 1' using the HYPOTHESIS_COMBINATION strategy. The debates revealed that both evaporation and surface effects were plausible, hardware-specific mechanisms. This hybrid model addresses the possibility that they are not mutually exclusive but are in fact synergistic, providing a more robust and comprehensive explanation for the observed yield loss. Its superior testability lies in a proposed 2x2 factorial experiment (Evaporation Control vs. Surface Passivation) designed to de-convolute these two effects. **Key Supporting Evidence:** - Rationale for winner 'Architect 1': Praised for its parsimony and proposing a direct, testable physical mechanism (evaporation) to explain a known biochemical failure. - Rationale for winner 'Explorer 1': Acknowledged for proposing a novel biophysical mechanism (surface adsorption) that directly links the hardware to the biochemical failure mode. - Finding: The accumulation of inorganic phosphate (P) is a primary inhibitor of protein synthesis because it sequesters free magnesium ions (Mg). (Literature) - Opportunity: There is a lack of research characterizing how physical reaction conditions (e.g., specific hardware, surface-to-volume ratios, evaporation) interact with and alter the optimal biochemical composition of the PURE system. (Literature)

2: Dynamic Stoichiometric Collapse: Evaporation as a Physical Trigger for Critical Biochemical Imbalance

Hypothesis: This hypothesis reframes the stoichiometric balance of the PURE system as a dynamically fragile state. It posits that the optimal window of component ratios (especially free Mg vs. ATP vs. Creatine Phosphate) is narrow. The accelerated evaporation on the Labcraft hardware acts as a continuous physical perturbation that actively drives the reaction out of this optimal window. By concentrating all solutes, evaporation specifically accelerates the sequestration of free Mg by inhibitory byproducts (P), leading to a 'dynamic collapse' of the system's stoichiometry. Therefore, the lower yield on Labcraft is not due to a poor starting recipe, but to a hardware-induced failure to *maintain* the required biochemical equilibrium throughout the reaction. **Final Rationale:** Evolved from 'Quant 1' using the HYPOTHESIS_COMBINATION strategy to address its primary weakness identified in debates: a lack of connection to the hardware-specific problem. By explicitly linking the concept of stoichiometric balance to the winning physical mechanism of evaporation (from 'Architect 1'), this hypothesis now provides a direct, testable explanation for platform-dependent variance. It moves beyond a static description of biochemical conditions to a dynamic model of hardware-induced failure, making it far more relevant to the core research question. **Key Supporting Evidence:** - Rationale for loser 'Quant 1': Criticized for describing the universal biochemical conditions of failure without explaining the platform-specific trigger. - Rationale for winner 'Architect 1': Identified a specific cause (evaporation) for the platform variance, providing a clear, actionable path to a solution. - Finding: The optimal total magnesium concentration is critically dependent on the concentrations of NTPs and inorganic phosphate due to strong chelation effects. (Literature)

Proposed Experimental Plan

Objective

To test the 'Synergistic Biophysical Failure' hypothesis by systematically and combinatorially evaluating the main and interaction effects of two key physical factors—evaporation and surface properties—on PURE system protein yield within the Labcraft hardware. This experiment is designed to de-convolute whether yield suppression is primarily driven by evaporation-induced concentration, surface-mediated sequestration, or a synergistic combination of both.

Multivariate Design Space

This experiment utilizes a 2x2 full factorial design for the physical variables, coupled with a combinatorial titration of key biochemical components within each factorial block. This creates a highly coupled, multi-dimensional experimental batch that avoids OFAT (One Factor At a Time) testing.

1. Physical Factorial Design (2x2): * **Factor A: Evaporation Control** * Level 1: No Control (Standard Labcraft operation) * Level 2: Control (Mineral oil overlay applied to each well post-setup) * **Factor B: Surface Passivation** * Level 1: No Passivation (Standard reaction plates) * Level 2: Passivation (Pre-treatment of wells with a 1% BSA solution, followed by washing, to block non-specific binding sites)

This design results in four distinct experimental blocks (e.g., a 96-well plate can be divided into four 24-well quadrants, one for each condition).

2. Biochemical Combinatorial Titration (within each block): Within each of the four physical condition blocks, a small, information-rich set of experiments will be run, simultaneously varying the most critical biochemical components identified in the 'Dynamic Stoichiometric Collapse' hypothesis.

- **Variable 1:** `[Magnesium acetate]`: Titrated across a range known to span suboptimal and optimal conditions (e.g., 6 mM, 12 mM, 20 mM).

- **Variable 2: `[Creatine phosphate]`**: Titrated across medium to high ranges (e.g., 20 mM, 40 mM, 60 mM).
- **Variable 3: `[ATP]` / `[GTP]` ratio**: While keeping the total nucleotide concentration constant, the ratio will be varied (e.g., ATP: 2.5 mM / GTP: 1.5 mM; ATP: 2.0 mM / GTP: 2.0 mM).

This biochemical titration will not be exhaustive but will create a grid of conditions designed to stress the system and reveal interactions with the physical factors.

Batch Protocol Steps

1. **Plate Preparation**: Prepare two 96-well plates. For the 'Passivation' plate, incubate all wells with 1% BSA for 30 minutes, then wash 3x with nuclease-free water and dry. The 'No Passivation' plate is used as-is.
2. **Master Mix Preparation**: Prepare a base master mix containing all PURE system components held constant (e.g., Pmix, Ribosomes, DNA, etc.).
3. **Automated Titration**: Using the Labcraft liquid handler, dispense the base master mix into all wells of both plates. Then, program the Labcraft to perform the combinatorial titration of Magnesium acetate, Creatine phosphate, ATP, and GTP according to the design space outlined above, creating identical biochemical grids on both plates.
4. **Evaporation Control**: For the designated 'Control' halves of each plate, carefully dispense a 15 µL layer of mineral oil on top of the 10 µL reaction in each well.
5. **Incubation & Measurement**: Place both plates in the reader and measure fluorescence (for deGFP expression) over time. Ensure plate reader settings are identical for both plates.

Optimization Metrics

- **Primary Metric**: `sigmoid_steady_state` (Final protein yield). The goal is to observe a significant increase in this metric in the Evaporation Control and/or Surface Passivation blocks compared to the No Control/No Passivation block.
- **Secondary Metric**: `sigmoid_rate (1/h)` (Rate of protein synthesis).
- **Analysis**: The results will be analyzed using ANOVA or a similar statistical method to determine the significance of the main effects (Evaporation, Passivation) and the interaction effect (Evaporation * Passivation).
  - A strong main effect of **Evaporation Control** would validate the 'Dynamic Stoichiometric Collapse' hypothesis.
  - A strong main effect of **Surface Passivation** would support the surface-adsorption component of the 'Synergistic Biophysical

Failure’ hypothesis.
    - A significant **interaction effect** would provide strong evidence for the full ‘Synergistic Biophysical Failure’ hypothesis, indicating that both mechanisms contribute and their combined negative impact is greater than the sum of their parts.
- **Feedback to Surrogate Model**: The entire dataset, comprising 4 distinct blocks of combinatorial biochemical data, will be fed back into the surrogate model. This will train the model on the impact of these physical variables, allowing it to predict optimal biochemical recipes that are robust to the specific hardware environment, thereby actively driving the optimization process toward our goal.

References

- [Improved Cell-Free RNA and Protein Synthesis System, Li J, Gu L, Aach J, Church GM, 2014, PLOS ONE]
- [Optimization of PURE system composition using automation and active learning, Vervel et al., 2026, bioRxiv]
- [Protein synthesis yield increased 72 times in the cell-free PURE system, Yellman and Jewett, 2016, Biotechnology and Bioengineering]
- [Cell-free translation reconstituted with purified components, Shimizu Y, Inoue A, Tomari Y, et al., 2001, Nature Biotechnology]
- [ATP Regeneration from Pyruvate in the PURE System, Lavrinenko et al., 2025, ACS Synthetic Biology]

Dead Ends

- Hypothesis: Stoichiometric Balancing of the Energy-Cofactor System Governs PURE Yield (Original ‘Quant 1’). Reason for rejection: While accurately describing the universal biochemical landscape of the PURE system, this hypothesis failed to propose a specific, testable mechanism to explain the observed hardware-dependent yield variability, which was the central research question. It described the ‘what’ of the failure state but not the platform-specific ‘why’.